# Seek-V2: Efficient MoE Language Model with Multi-Head Latent Atte

236B total parameters / 21B activated per token

128K-token context length

Pretrained on 8.1T tokens

Source: DeepSeek-V2 (arXiv:2405.04434)

# The LLM Scaling Dilemma: Performance vs. Cost and Latency

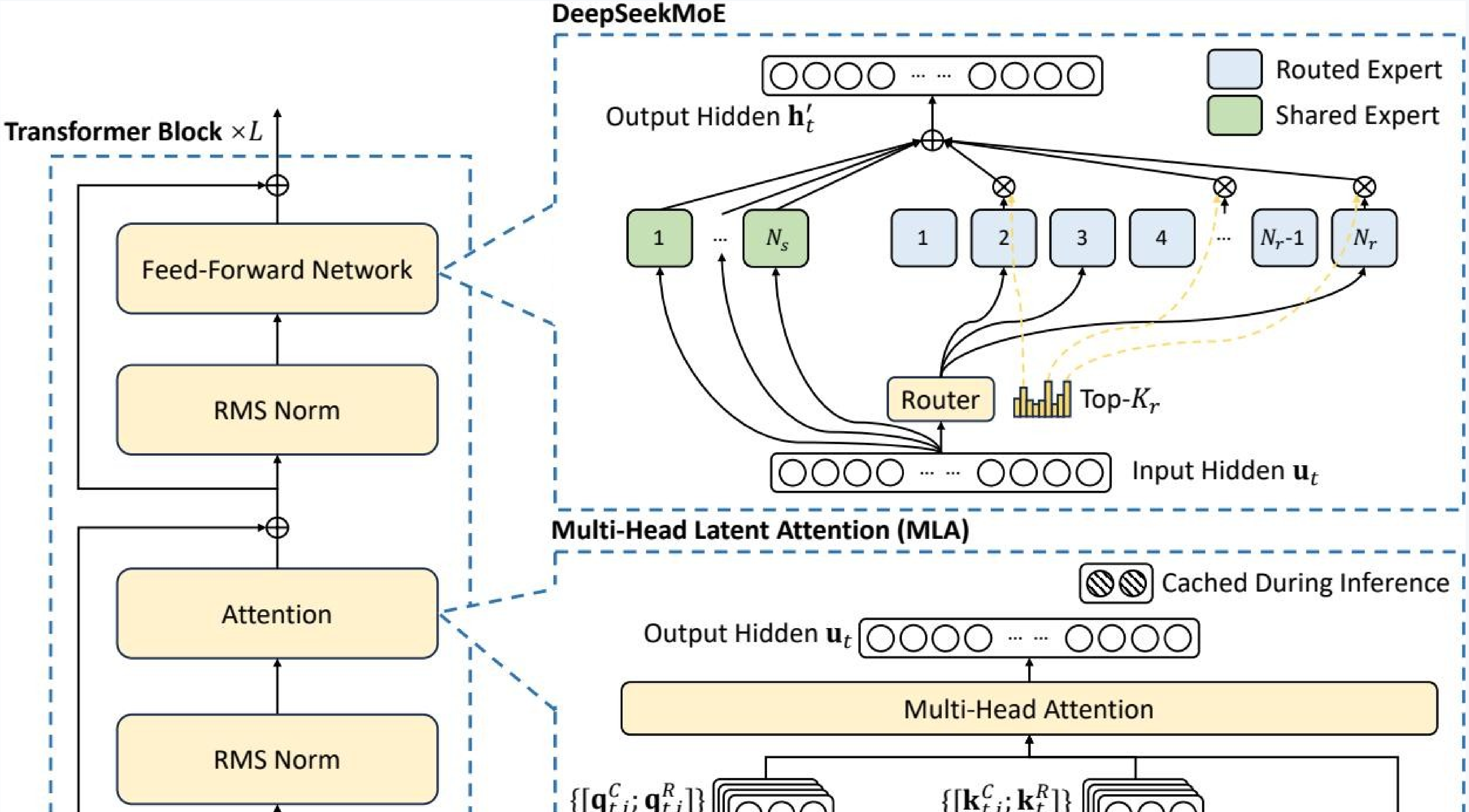
Scaling dense LLMs improves capability but sharply increases training cost

Autoregressive decoding is constrained by KV cache memory bandwidth

GQA/MQA reduce cache pressure but can degrade quality

Need: architecture that jointly improves quality, cost, and throughput

# DeepSeek-V2 Architecture: MLA + DeepSeekMoE



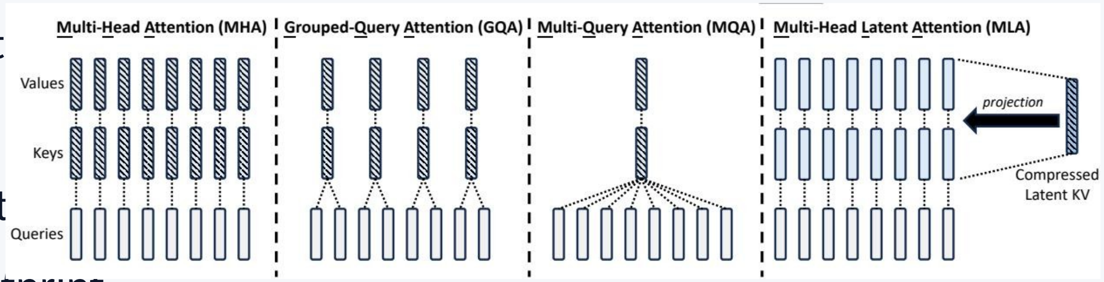
# MLA: Multi-Head Latent Attention

Low-rank key-value joint compression caches compact latent vector c_t

Full K/V and decoupled RoPE positional key reconstructed on-the-fly

Reduces inference-time KV cache while preserving attention expressivit

Reported to outperform conventional MHA with much lower cache footprint



# DeepSeekMoE: Fine-Grained Experts + Shared Isolation

Shared experts (N_s): always active; capture common knowledge

Routed experts (N_r): sparse top-K_r routing per token

Fine-grained expert segmentation improves specialization

Device-limited routing reduces cross-node communication

Auxiliary losses + token dropping stabilize load balancing

# Efficiency Gains vs. DeepSeek 67B

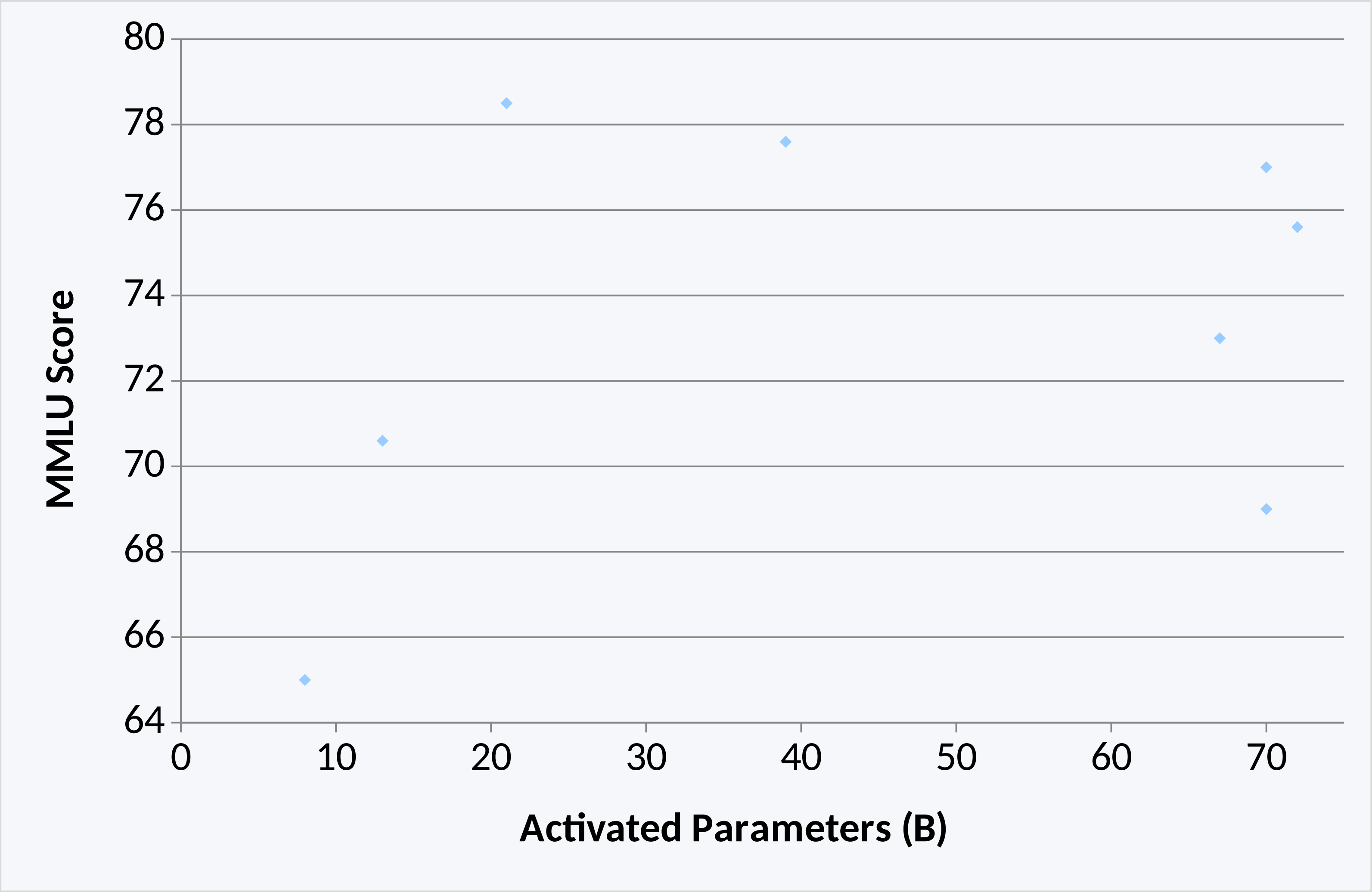
## DeepSeek 67B

| | |
|---|---|
| Training Cost | Baseline |
| KV Cache Size | Baseline |
| Max Generation Throughput | Baseline |

## DeepSeek-V2

| | |
|---|---|
| Training Cost | 42.5% lower |
| KV Cache Size | 93.3% lower |
| Max Generation Throughput | 5.76× higher |

# MMLU vs. Activated Parameters: Pareto-Optimal Positioning




DeepSeek-V2: (21B, 78.5)
LLaMA 3 70B: (70B, 77.0)
Mixtral 8x22B: (39B, 77.6)
Qwen1.5 72B: (72B, 75.6)
DeepSeek 67B: (67B, 73.0)
LLaMA 2 70B: (70B, 69.0)
Mixtral 8x7B: (13B, 70.6)
LLaMA 3 8B: (8B, 65.0)

# Alignment Results: DeepSeek-V2 Chat (RL)

Post-training: SFT on 1.5M conversational sessions + GRPO RL

AlpacaEval 2.0 (length-controlled win rate): 38.9%

MT-Bench (overall): 8.97

AlignBench (overall, Chinese): 7.91

Outperforms all open-source and most closed-source models on AlignBench

# Key Takeaways

MLA is a high-quality, low-cache attention mechanism (93.3% KV reduction)

DeepSeekMoE enables 236B total scale with only 21B activated parameters

DeepSeek-V2 cuts training cost by 42.5% vs. DeepSeek 67B

Throughput gain of 5.76× improves real-world deployment efficiency

Model sits on a strong cost-performance Pareto frontier among open models